Elijah Ifechukwu Nelson

Machine Learning & Embedded Systems Engineer

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Summary

Machine Learning and Embedded Systems engineer with an M.Phil. in Machine Learning from the University of Cambridge and a valedictorian in Electrical & Electronics Engineering. Builds systems end-to-end, from low-power embedded firmware and custom sensor hardware to production ML retrieval pipelines, with 1,000+ field instruments deployed across four countries and published research in robotic manipulation. Focused on the intersection of ML and Systems, particularly efficient methods for resource-constrained deployment.

Education

M.Phil. in Machine Learning and Machine Intelligence, University of Cambridge, Cambridge, UK — 10/2025 - 08/2026

- Advanced coursework in Probabilistic Machine Learning, Optimization, and Reinforcement Learning.
- MPhil thesis, supervised by Andrew Fitzgibbon (formerly Microsoft Research): investigating whether structured (Kronecker-factored, low-rank) approximations of the transformer Hessian yield optimisers that match or exceed Muon, Shampoo, and SOAP on language-model pretraining. Includes from-scratch implementations of Muon and Shampoo, full-Hessian and Gauss–Newton spectral analysis on small transformers, and scaling-law evaluation under the Wen et al. (2025) fair-comparison protocol.

B.Eng. in Electrical and Electronics Engineering, Covenant University, Ota, Nigeria — 08/2018 - 09/2023

- First Class Honours; CGPA 4.98/5.0, the highest in the department's history.
- Class rank 1 of 132; institute rank 1 of 1,175 (Valedictorian).
- Thesis: designed, simulated, and built a multi-purpose autonomous UAV with autonomous navigation, precision landing, and visual tracking. [link]

Research Interests

ML + Systems, Embedded AI, Embedded Systems, Sensor Design, IoT, Embedded Software & OS, Computer Architecture, Robotics, Integrated Circuits.

Research & Teaching Experience

AI Research Assistant, Intelligent Manipulation Lab, University of Lincoln, Lincoln, UK — 03/2022 - 06/2023

- Developed a novel SMOTE-CNN technique that classifies slip in real time from temporal tactile data without relying on kinematic state, achieving 98% slip-detection accuracy and a 95% reduction in false negatives on a limited, imbalanced dataset. [link]
- Conducted experiments comparing AI ensemble models and tactile data representations to advance slip classification in robotic manipulation.
- Assessed SHAP and saliency-based explainability methods, then applied Grad-CAM to 16×16 uSkin tactile images across 16 test trials, revealing that edge and central taxels with significant shear-force changes drove SMOTE-CNN slip predictions.

Research Assistant, Dept. of Electrical & Electronics Engineering, Covenant University, Ota, Nigeria — 01/2023 - 07/2023

- Built embedded firmware on the ATmega328-PU for two autonomous/IoT systems: an automated waste-management unit and an autonomous vacuum-cleaning robot.
- Waste unit: used ultrasonic sensors with interrupt-driven wake-up and a low-power SIM800L module for SMS notifications, achieving reliable operation across all tested configurations through careful power management.

- Vacuum robot: implemented ROS-based autonomous navigation with a laptop master node running the core navigation and computer-vision algorithms, consuming odometry and issuing control commands over ROS nodes; simulated in Gazebo and teleoperated via a game controller.
- Taught and trained 50+ students in embedded-systems research methodologies.

Tutor, Covenant University, Ota, Nigeria 01/2021 - 07/2023

- Taught 200+ students across Electromagnetism, Applied Electronics, Advanced Control Systems (ACS), Power Electronics, and High Voltage Engineering, using personally curated notes and interactive case studies.
- Achieved an 85% average pass rate across subjects, including 77% in ACS.

Publications

[1] N. Elijah, K. Nazari, and A. Ghalamzam. Advancing slip classification in robotic manipulation through tactile data representation and model selection. In *IEEE Sensors Journal*, 2025. [link].

[2] T. Somefun, N. Elijah, A. E. Igho-Orere, O. Timilehin, C. Somefun, and S. Ongbali. Energy optimization algorithm for reducing energy consumption in a smart home. In *2023 2nd International Conference on Multidisciplinary Engineering and Applied Science (ICMEAS)*, pages 1–5, 2023. [link].

Professional Experience

Founding Systems Engineer, Climate in Africa, Lagos, Nigeria 10/2023 - 08/2025

- Built low-cost, WMO-compliant Modular In-situ Reporting Instruments (MIRI) measuring 14 meteorological parameters, with 1,000+ units deployed across Nigeria, Ivory Coast, Kenya, and Ghana.
- Wrote a NOR Flash (MT25QL01GBBB) driver and co-designed CRC-verified OTA updates over BLE and octet streams for resilient field upgrades; ported 3K LOC from ATmega to STM32 (F4/F103) for a 30% performance gain.
- Designed MIRI firmware reaching 3.5 days per charge using a sleep scheduler, an RTOS, and non-blocking state machines, cutting GSM/GPRS latency by ~70%, with autonomous fallback logging to NOR Flash/SD card on connectivity loss.
- Built the mail and notification architecture for climateinafrica.com, aggregating TBs of data from MIRI monitors and users, and architected a NOAA pipeline to scrape and parse GB-scale climate datasets for the analytics sandbox.
- Led the entire product documentation effort for climateinafrica.com.
- Developed a low-earth orbit (LEO) satellite tracker using non-blocking task scheduler for autonomous satellite tracking (azimuth and elevation actuation) and a self-designed LoRa base-to-ground-station link for telemetry and control.
- Performed testing and validation of UHF/VHF satellite radios using spectrum and logic analysers.

Machine Learning Engineer, Ticki, Remote 08/2024 - 07/2025

- Architected a facial-similarity retrieval system (OpenCV, FAISS, async Celery) serving 5,000 users, cutting end-to-end latency by ~70% to sub-100 ms by migrating the index from FAISS IndexFlatL2 to IndexIVFFlat and caching face embeddings in memory.
- Built the core pipeline: on first run, extract every face in the database and store its embedding as a NumPy array keyed to the source filename for traceability; on query, extract the input face and match it against the embedding index.
- Handled multi-face queries with an interactive selection step backed by in-memory BytesIO buffers instead of disk; offloaded compute-intensive face-extraction and embedding to Celery workers, using Redis to deduplicate already-processed images and avoid redundant recomputation across a ~3,000-image dataset.
- Hardened a production Ubuntu VPS deployment for near-zero downtime under 1,000+ concurrent requests: a UFW firewall restricting access to designated IPs (no 0.0.0.0 exposure), scoped user groups, multiple Gunicorn workers, and Nginx load balancing.

Reservoir Performance Simulation Intern, Schlumberger (SLB), Port Harcourt, Nigeria 04/2022 - 08/2022

- Performed rock and fluid analysis on core samples in the wet lab (rheology, density/PVT, calorimetry, and precision measurements) to characterise formation stability parameters for drilling; interpreted subsurface and wellbore data in Petrel and Techlog.
- Maintained and tested mechanical systems (CPS-361 surface units, compressors, generators) against full pre-deployment maintenance checklists before shipment to wellsites: sensor testing (flow, pressure, temperature, optical), oil changes, and repairs; raised maintenance throughput by ~15% and proposed an automated CPS-361 design.
- Managed IT and networking for the entire SLB base: resolved issues across laptops, desktops, printers, and routers, configured access points and new-employee machines, and handled the IT ticket queue.

Delegate of Croatia, Model United Nations, Beirut, Lebanon 03/2019 - 04/2019

- Researched Indigenous cultural heritage and the preservation of cultural artifacts; prepared committee reports for UNESCO.

Delegate of Israel, Model United Nations, Buenos Aires, Argentina 09/2018 - 10/2018

- Researched policies for mitigating global climate change in developing countries.

Selected Projects

CamAI: Action Recognition Camera System, MediaPipe, Conv-LSTM, Computer Vision 03/2023 - 05/2023

- Led a team of two to build an action-recognition system for patient monitoring in healthcare.
- Extracted pose features from the video feed with MediaPipe and classified patient state with a Conv-LSTM ensemble at 96%+ accuracy, with reliable subject tracking, streamed to a web interface.

Quad-X: Quadcopter Flight Controller, ATmega328-PU, PID, Sensor Fusion 01/2021 - 12/2021

- Designed and built a quadcopter flight controller on an ATmega328-PU, fusing IMU data through a PID control loop for stable all-axis flight.
- Extended Quad-X into AutoUAV, adding autonomous navigation, in-flight object detection, and live camera streaming.

MilliMeter: Low-Cost IoT Metering System, Embedded Communication Protocols, Web 03/2020 - 07/2020

- Led a team of 4 to build a low-cost IoT smart-metering system using custom embedded communication protocols and web technologies for a Lagos State smart-meter competition.

Skills

- **Programming Languages:** Python, C, C++, JavaScript, SQL.
- **AI & Deep Learning:** PyTorch, TensorFlow, scikit-learn, NumPy, Matplotlib, seaborn, FAISS, OpenCV.
- **Frameworks & Tooling:** Flask, Celery, Redis, Gunicorn, Nginx, ROS, Gazebo.
- **Embedded & Hardware:** STM32, ATmega, microcontrollers, RTOS, I2C, SPI, UART, BLE, LoRa, GSM/GPRS, sensors, soldering, KiCad.
- **DevOps & Version Control:** Git, GitHub, Linux/Ubuntu (VPS, UFW).
- **Languages:** English (native).
- **Soft Skills:** Technical writing; teaching; project, people, and time management; leadership; critical thinking.

Training & Certifications

- **Deep Learning Specialization**, DeepLearning.AI (Coursera).
- **Computer Architecture Essentials on Arm**, ArmEducationX.

Leadership Experience

Administrative Head, Luminous Life Foundation, Lagos, Nigeria 09/2023 - Present

- Led a team of 15 to deliver solar power to 100 homes (∼400 residents) in rural Nigerian communities that had been without electricity for over 50 years.

Academic Officer, Dept. of Electrical and Electronics Engineering, Covenant University, Ota, Nigeria 09/2022 - 08/2023

- Represented 500+ students to 20 faculty members as the department's academic liaison.
- Coordinated tutorials and curated past-question banks with TAG for students across all years.
- Organized symposiums connecting Electrical Engineering students with industry leaders.

Volunteer Work & Community Service

Co-founder, Tech for Progress, Isolo, Lagos, Nigeria 01/2023 - Present

- Launched Isolo's first digital-skills training for underserved youth, equipping 50+ secondary-school students with Microsoft Office (Word, PowerPoint, Excel) and design (Canva) skills.
- Scaling into Phase 2 by training teachers alongside students for continuity, with several students now progressing into bootcamps.

Career Advisor, Tesla's Academic Group (TAG), Covenant University, Ota, Nigeria 03/2022 - 03/2023

- Guided 400+ students in their academic and career development through periodic webinars.

Lead AI Tutor, Google Developer Student Committee (GDSC), Covenant University, Ota, Nigeria 01/2022 - 12/2022

- Grew the AI community from 200 to 500+ members through a custom curriculum.
- Led the community in building NLP and computer-vision (image-detection) projects.

Member / Assistant Project Manager (National Champions), Enactus, Covenant University, Ota, Nigeria 11/2019 - 09/2022

- Led development of ECOBAG, a heat-retention cooler helping SMEs cut spoilage under unstable power supply.
- Co-developed PET-City, a plastics-to-bricks scheme to reduce plastic waste.
- Earned National Champions status among 18 teams nationwide (2020) for PET-City's impact.

Tutor, Deeper Life Student Outreach (DLSO), Ojo, Lagos, Nigeria 07/2019 - 08/2019

- Prepared 20+ students for college entrance exams and advised on their career choices.

Honors & Awards

- **Fitzcelerate**, University of Cambridge, £2500 – Runner-Up (2026).
- **Charitable and Community Awards**, Fitzwilliam College, University of Cambridge (2026).
- **Mastercard Foundation Scholar**, University of Cambridge – 69 of 4,218 (2025).
- **Best Graduating Student**, Covenant University – 1 of 1,175 (2023).
- **Best Graduating Student in Engineering & EIE**, Covenant University – 1 of 132 (2023).
- **Highest CGPA (4.98/5.0)** in the EIE Department's history at Covenant University (2023).
- **Dean's Award for Academic Excellence**, Covenant University – 1 of 386 (2023).
- **Femi Ademiju Award for Excellence** to the Best Graduating EIE Student (2023).
- **CUALA Award for Best Graduating Student**, College of Engineering, Covenant University (2023).
- **First Runner-Up**, 3-line Finance Hackathon (2022–2023).
- **First Runner-Up**, Innovation Hackathon, Covenant University (2022).
- **$100 Prize**, OMGRobotics Challenge, Goodwall (2021).
- **$250 Prize**, My Passion Challenge, Goodwall (2021).
- **Qualifier**, Lagos State Smart Meter Hackathon (2020).
- **Winner**, Thinkerthon (2020).
- **First Runner-Up**, Pan-African Hackathon (2020).
- **Best Student in IGCSE** and in Chemistry & Geography Studies, DLHS, Lagos (2018).

Media Appearances

- Best Graduating Student in Covenant University – featured (09/2023).
- 3line's Hackathon – featured (04/2023).

Interests

Piano, cycling, running, reading, writing, blogging, coaching, and coding for fun.